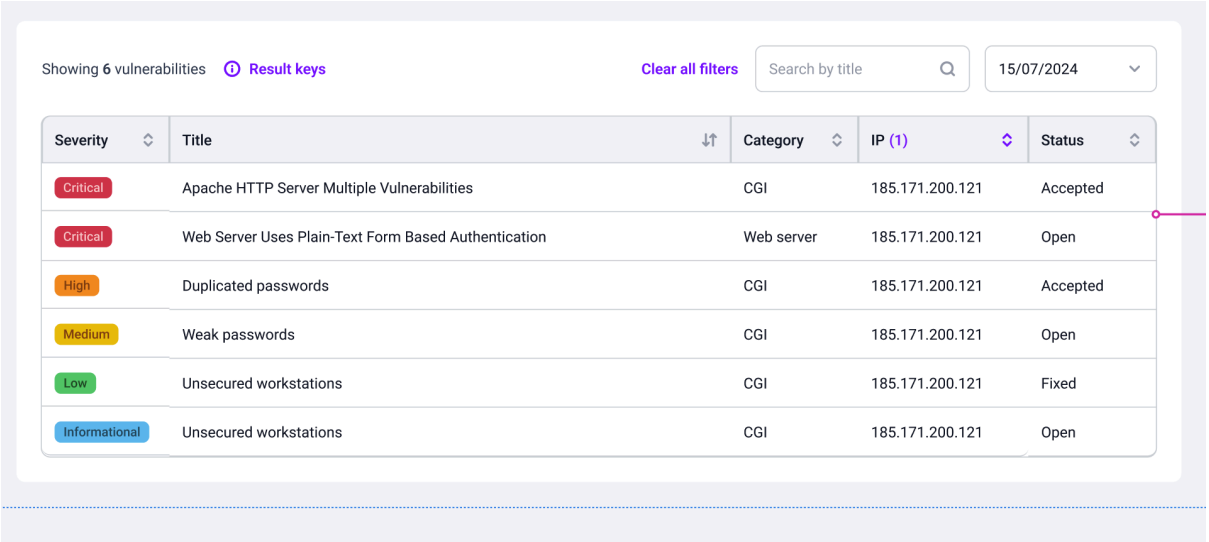
Assignment: Full-Stack Vulnerability Management Dashboard



Overview

You are required to develop a vulnerability management dashboard that displays security vulnerabilities with different severity levels, categories, IP addresses, and statuses. The application should allow filtering, sorting, and searching vulnerabilities while ensuring data persistence and real-time updates.

Backend (BE) Requirements

Tech Stack: PostgreSQL, GraphQL API

1. **Database Schema & Models**
    - Design a database schema to store vulnerability reports with fields as shown in the table
    - We are expecting that a user have a Vulnerability Scanner project and for that project we need to pull the vulnerability results from the table
2. **API Endpoints**
    - **GET /vulnerabilities** – Fetch all vulnerabilities with filters (`severity`, `status`, `category`).
    - **Authentication & Authorization**
        - API should be accessible if authenticated.
        - Implement role-based access control (Admin, Viewer).

Frontend (FE) Requirements

Tech Stack: Next.js, React, TailwindCSS, GraphQL

1. **Dashboard UI**
    - Display vulnerability data in a table format with columns:
        - **Severity** (Color-coded: Red = Critical, Orange = High, Yellow = Medium, Green = Low, Blue = Informational)
        - **Title**
        - **Category**
        - **IP Address**
        - **Status** (Dropdown to update)
    - Implement sorting by severity, category, and status.
2. **Filters & Search**
    - Provide a search bar to search by title.
    - Filters for severity, status, and category.
    - Date filter to select vulnerabilities within a range.

Nice to have

- UI implementation as per the given design
- Unit test case